

Nathan Spelts

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SUMMARY

I write embedded C for automotive ECU reprogramming, after nearly three years on safety-critical forklift firmware, secure boot, and Rust tooling for CAN diagnostics. Currently pursuing an M.S. in Cybersecurity at Georgia Tech, focused on cyber-physical systems.

WORK EXPERIENCE

Salvo Software, Vancouver, WA

Apr 2026 — Present

- Firmware Engineer

- Maintain production embedded C firmware for an automotive diagnostic tool that reprograms vehicle ECUs.
- Implement flash routines for reprogramming a wide range of ECUs, with a focus on reliability.
- Develop .NET cloud services that support the diagnostic tool.

Hyster-Yale Materials Handling, Fairview, OR

Aug 2023 — Apr 2026

- Software Engineer II

- Developed safety-critical embedded C firmware for forklift controllers under industry safety standards.
- Improved secure boot and firmware authentication to guard against tampering and unsafe OTA updates.
- Built a Rust CAN logger and analyzer for Vector XL hardware via C FFI, for live and offline diagnostics.

Seasalt.AI, Remote

Jul 2022 — May 2023

- Data Science Intern

- Reduced word error rate over 40% for Spanish and Indonesian Kaldi ASR models through model and data tuning.
- Automated scraping and cleaning of thousands of hours of captioned training audio in Python and Bash.

EDUCATION

Georgia Institute of Technology, Remote

Aug 2025 — Present

- Master of Science in Cybersecurity – *Cyber-Physical Systems*

- Current GPA: 4.00
- Coursework: secure system design, binary exploitation, network and database security, applied cryptography.

Washington State University, Vancouver, WA

Graduated May 2023

- Bachelor of Science in Computer Science

- GPA: 3.9 / 4.00

PROJECTS

Raspberry Pi 4 Bare-Metal Kernel

github.com/micro-wizard/raspi4_rust_bootloader

- A no_std Rust kernel for the Raspberry Pi 4, booted straight from the SD card.

- Wrote AArch64 startup assembly that parks secondary cores, zeroes BSS and sets up the stack for Rust.
- Drove the VideoCore GPU via its mailbox interface for a double-buffered framebuffer with text rendering.
- Unit-tested the hardware-facing code on the host against a mocked mailbox, run in GitHub Actions CI.

SKILLS

Programming Languages: C, C++, Rust, Python, C#, Bash

Embedded & Systems: Secure Boot, Bootloaders, Bare-Metal ARM, RTOS, I2C / SPI / UART / CAN, Memory-Mapped I/O, GDB, QEMU

Tools & Technologies: .NET, Git, GitHub Actions, Docker, Linux, FFI (Rust/C), Vector XL